



Open-source Rietveld refinement for cellulose crystallinity using a laboratory powder X-ray diffractometer: a hands-on tutorial

Sunghyun Nam · Alfred D. French ·
Md Nayeem Hasan Kashem

Received: 22 August 2025 / Accepted: 30 June 2026

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Abstract Accurate measurement of cellulose crystallinity is essential for understanding fiber development, optimizing processing conditions, and improving material performance. However, the structural complexity of cellulose fibers presents significant challenges in determining crystallinity, or more precisely, the ratio of two discrete crystalline and amorphous model phases. As a result, many studies rely on empirical methods to estimate a crystallinity index (CrI). This study presents a cellulose-specific workflow for quantifying crystallinity from one-dimensional X-ray powder diffraction (XRD) data using Rietveld refinement implemented in the open-source software MAUD. The step-by-step procedure—demonstrated using cotton fiber and tailored

for non-specialists—guides users through refinement of ten key parameters, including a scale factor, background, unit cell geometry, crystallite size, preferred orientation, and phase composition, in four sequential refinement steps. The final refinement yielded strong agreement between calculated and observed diffraction patterns and demonstrated the utility of one-dimensional full-pattern analysis in a workflow to quantify cellulose crystallinity.

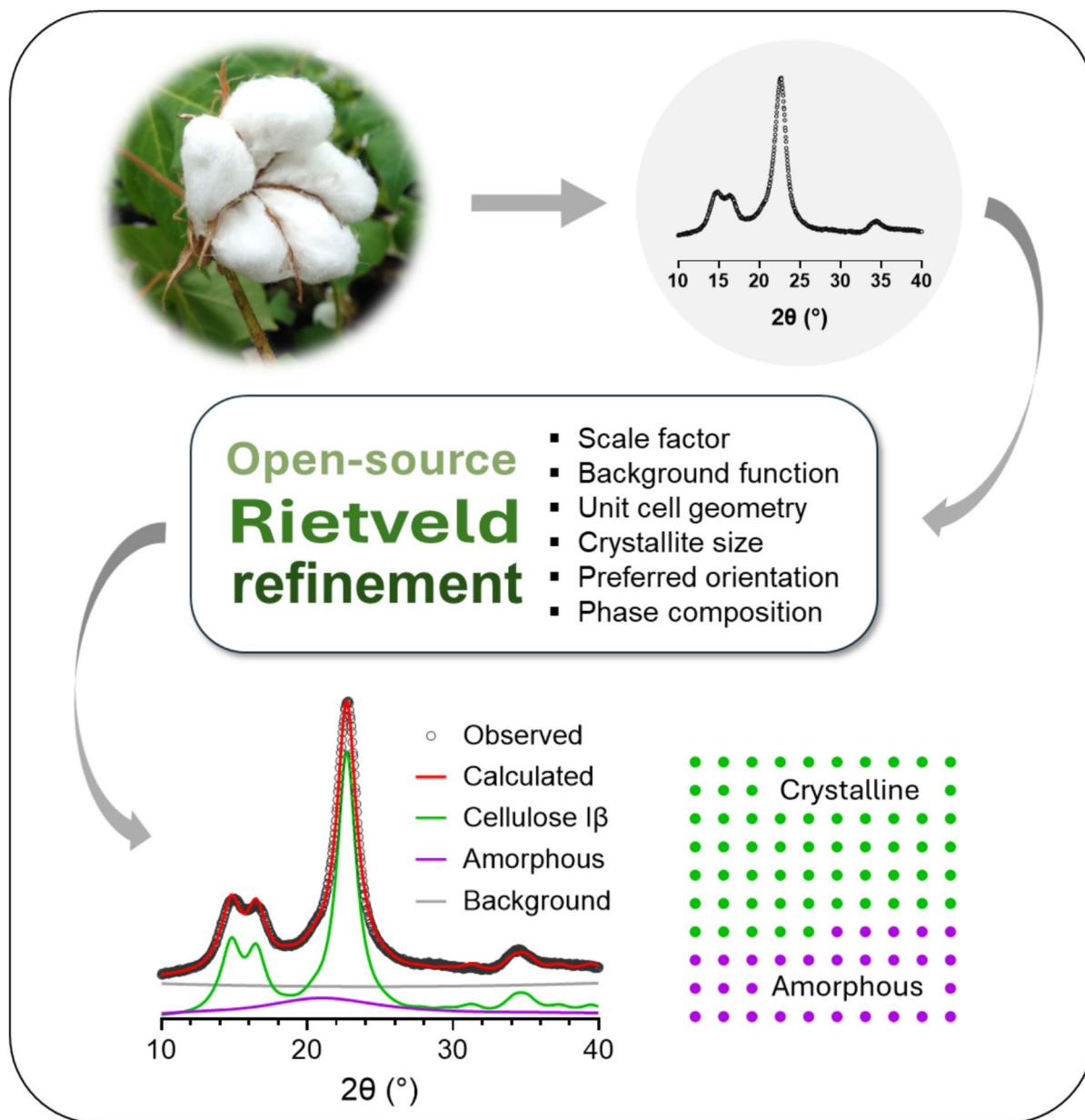
Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10570-026-07136-7>.

Alfred D. French: Retired.

S. Nam (✉) · Md. N. H. Kashem
Cotton Fiber Bioscience and Utilization Research Unit,
Southern Regional Research Center, U.S. Department
of Agriculture, Agricultural Research Service,
New Orleans, LA 70124, USA
e-mail: sunghyun.nam@usda.gov

A. D. French
Southern Regional Research Center, U.S. Department
of Agriculture, Agricultural Research Service,
New Orleans, LA 70124, USA

Graphical Abstract



Keywords Determination of crystallinity · XRD · Rietveld refinement · Amorphous cellulose · Preferred orientation · Cotton fibers · MAUD

Introduction

Cellulose fibers, in both naturally occurring and regenerated forms, play essential roles in industries ranging from textiles to composites and bio-based materials (Ramamoorthy et al. 2015; Wang et al. 2021). Their performance characteristics, such as

strength, flexibility, chemical reactivity, and moisture management, are closely tied to their structural organization at the molecular level (Ward Jr 1950). Cellulose is commonly described as a semi-crystalline polymer composed of crystalline and amorphous regions (Wakelyn et al. 2006). In crystalline domains, cellulose chains are tightly packed and aligned in parallel, whereas in amorphous regions, they are disordered. Molecular modeling studies, on the other hand, suggest that the nanoscale organization of cellulose may also include intermediate, partially ordered regions that are not cleanly classified as either fully crystalline or fully amorphous (Haigler et al. 2014; Kulasinski et al. 2014). These paracrystalline regions were found to exhibit intermediate mechanical properties between crystalline and amorphous phases (Kulasinski et al. 2014). From a practical viewpoint, a two-phase model is possibly the closest approximation to the real situation that can be obtained. Fink et al. (1995) showed for example that existing lattice strains have minimal effect on crystallite size measurements. Nishiyama et al. (2012) showed that larger strains resulting from molecular dynamics-based distortion had small effects on calculated diffraction patterns.

During biosynthesis in the developing fiber, cellulose crystallizes predominantly in cellulose I allomorph (Fang and Percy 2020), though it can be converted into cellulose II, III, or IV through post-treatment processes, such as mercerization or thermal/chemical modification (Colom and Carrillo 2002; Szabó et al. 2023). Crystallite size, which varies with fiber type, development stage, and treatment history, influences the specific surface area of cellulose crystallites, thereby affecting their chemical reactivity and interaction with other molecules. Meanwhile, the amorphous phase contributes to diffusion properties and flexibility, often serving as a target for chemical and enzymatic modifications (Emenike et al. 2023). Because the balance between these two phases governs many of the fiber's functional properties, the degree of crystallization, or crystallinity, is an important metric in cellulose fiber science, widely used to evaluate fiber development, processing conditions, and application performance.

The crystallinity of cellulose can be assessed using diffraction techniques that employ sub-nanometer wavelength radiation, such as X-rays, neutrons, or electrons in electron microscopy. Among these, neutron and electron-based techniques are less commonly

employed: neutron sources are typically limited to national laboratories, and electron beams can rapidly degrade cellulose samples. In contrast, X-ray diffraction (XRD) using powder diffractometers is widely accessible and offers a relatively low-cost, non-destructive approach for examining cellulose crystallinity. While diffraction remains the primary method for evaluating crystallinity, alternative techniques such as infrared (IR) and nuclear magnetic resonance (NMR) spectroscopy are frequently employed to distinguish between cellulose polymorphs (Lee et al. 2016). However, spectroscopic methods do not detect the long-range atomic order required for diffraction-based analysis. Instead, they imply crystallinity and allomorph type from the orientations or interactions of specific functional groups associated with a given crystalline form of cellulose.

Most traditional XRD studies on cellulosic materials have focused on determining a crystallinity index (CrI), which estimates the relative proportion of crystalline cellulose, with the remainder considered amorphous. Among these empirical approaches, the method of Segal et al. (1959), which is based on peak height measurements, is the most widely used due to its simplicity. However, the foundation of this method is flawed: it presumes that the intensity minimum near $18^\circ 2\theta$ in a cellulose I β pattern (recorded using CuK α radiation) is the result of amorphous cellulose. Later studies on the shapes of diffraction peaks revealed that much of the intensity in this region is the result of the overlap of large adjacent peaks, together with contributions from smaller peaks (French and Santiago Cintrón 2013; Nam et al. 2016). This effect can be readily demonstrated by simulating diffraction patterns of cellulose I β from a crystal information file (cif), which contains published atomic coordinates of its asymmetric unit (Nishiyama et al. 2002; French and Santiago Cintrón 2013) (Fig. 1A). Varying the full width at half maximum (FWHM) in the simulations highlights the extent of peak overlap (Fig. 1B). The Segal method fails to account for the full diffraction pattern and important factors such as crystallite orientation and crystallite shape variations, leading to inaccurate assessments of cellulose crystallinity (French 2022).

In the present study, we introduce a method to quantify crystallinity through Rietveld refinement with modifications specifically adapted for cellulose materials. Rietveld refinement is a full-pattern fitting approach that calculates XRD diffraction intensities

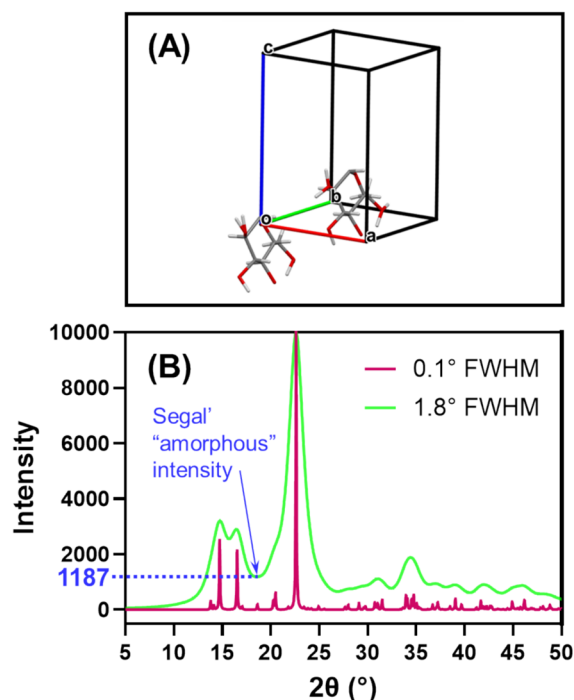


Fig. 1 **A** Asymmetric unit of cellulose I β from crystal information file (cif). **B** Calculated diffraction patterns for full width at half maximum (FWHM) of 0.1° (red) and 1.8° (green) using the Mercury software. These widths correspond to crystallite sizes of approximately 900 Å and 50 Å, respectively. The intensity at 18.6° on the 900 Å pattern, is essentially 0, so the Segal CrI is 100%. For the 50 Å pattern, the value of the intensity at the minimum near 18° is 1187, so the Segal crystallinity index (CrI) would be $100 \times (10,000 - 1187) / 10,000 = 88.1\%$ and the amorphous content would be 11.9%, even though both patterns were calculated for perfectly crystalline samples

from known crystal structures and iteratively adjusts structural and microstructural parameters of the known model structure to minimize the difference between the calculated and measured diffraction patterns. The method enables extraction of structural information from overlapping peaks, incorporates physically meaningful parameters, and can model complex diffraction behavior. For cellulose materials, however, several intrinsic features complicate the full application of conventional Rietveld analysis. Cellulose exhibits small crystallite sizes, extensive peak overlap, and a substantial amorphous contribution, and its unit cell dimensions, a , b , and γ vary considerably. Under such conditions, the effectiveness of refinement becomes influenced by how the

amorphous scattering is represented (Salem et al. 2023). In addition, some standard Rietveld implementations do not fully accommodate characteristic cellulose features, such as its needle-like morphology with an elongated c -axis along the molecular chain and shorter transverse dimensions.

The method introduced here is based on a two-phase representation of cellulose consisting of crystalline and amorphous cellulose components. To model this amorphous contribution, we prepared amorphous cellulose through mechanical treatment by high-impact ball-milling. The diffraction pattern from the ball-milled sample was then fitted with a broad-peak version of the cellulose II calculated structure. Cellulose samples from different sources and treatment histories will have different unit cell dimensions. We chose an FWHM of 9° corresponding to a cellulose II crystallite size of 12 Å as a representative of amorphous material. This pattern was a better fit than the wide peak versions of cellulose I. Apparently, the action of the impact ball mill destroyed the particular O2-H $\cdots$ O6 inter-residue intramolecular hydrogen bonds, allowing the O6 atoms to take the gt O6 orientation that is characteristic of cellulose II and amorphous cellulose. Another point mentioned in Plaza et al. (2026) is that if the isotropic crystallite size of the amorphous contribution is refined, it can exceed the limits for amorphous material. If the crystallite size exceeds 20 Å, the sample may be considered to be a mixture that includes crystalline cellulose II. If the refined crystallite size is less than about 8 Å, the resulting calculated intensity is just compensating for a deficit in the background correction. In this work, the crystallite size is kept fixed at 12 Å. Therefore the procedure makes use of the usual Rietveld provisions to allow flexible unit cell dimensions. The approach also incorporates accounting for preferred orientation—a departure from random orientation that is assumed in powder diffraction but extremely difficult to avoid in cellulose samples.

Here we present a step-by-step protocol for a cellulose-specific Rietveld refinement workflow using the freely available MAUD (Materials Analysis Using Diffraction) software. The procedure is demonstrated using cotton fibers as a representative example. This refinement approach has also been used effectively with other cellulosic materials including cellulose nanofibers (Easson et al. 2024; Nam et al. 2024a, b). The protocol outlines practical guidance for

non-specialists on selecting and adjusting key parameters, minimizing preferred orientation effects, and obtaining a structure-based measure of crystallinity. Preferred orientation was handled using the March-Dollase function, which empirically adjusts reflection intensities relative to a randomly oriented specimen. This workflow is designed to address common challenges in cellulose crystallinity analysis and to promote fuller use of powder diffraction data beyond qualitative visual inspection, pattern comparison, or simple peak-based metrics. Quantifying crystallinity through one-dimensional full-pattern fitting provides a basis for improving fiber development, process optimization, and the evaluation of material performance.

MAUD Rietveld software has many powerful capabilities. One of its strengths is provision for studying texture, the degree of preferred orientation. Rietveld software in general, including MAUD, also can take the place of a classical single-crystal diffraction study for determining the atomic coordinates of powdered samples, especially minerals of simple composition but unable to form crystals large enough for neutron diffraction. However, it is absolutely inappropriate to utilize any Rietveld software to attempt to carry out such a structure determination for cellulose structure with powder diffraction data, as there is simply not enough data to support determination of so many variables.

Materials and methods

Materials

Cotton fibers were obtained from cotton plants grown at the Southern Regional Research Center, New Orleans, LA. Methyl methacrylate, butyl methacrylate, benzoyl peroxide, and methyl ethyl ketone were purchased from Sigma-Aldrich (St. Louis, MO, USA) and used as received.

Sample preparation

To prepare samples for XRD measurement, harvested cotton fibers were mechanically ground using a sharp Wiley cutting mill (Arthur H. Thomas Co., Philadelphia, PA) fitted with a 40-mesh screen. The amorphous cellulose was prepared by further processing the Wiley-milled powder (approximately

5 g) in a custom-built vibratory ball mill operating at 1,750 rpm, following established procedures (Forziati et al. 1950). The milling was performed in a cylindrical steel jar of 1-L capacity, filled with approximately 500 mL of stainless-steel balls (6.35 mm diameter) as the grinding medium, and the ball-milling duration was 2h (Nam et al. 2020). Limitations include experimental handling of the background data as both a blank run and a secondary input data file in what Morales et al. (2026) call combined analysis. The default parameters were used in MAUD without specification of factors that vary from the standard setup such as variable slits instead of fixed slits and goniometer radius. Mentioned previously were the Caglioti parameters that were left in the default state. Advanced users should take these factors into consideration.

For cross-section preparation, the ground fibers were embedded in a mixture of methyl methacrylate, butyl methacrylate and benzoyl peroxide and polymerized using a UV cross-linker (UVP CL-1000, Analytik Jena) following published methods (Thibodeaux and Evans 1986; Boylston et al. 1991). The polymerized blocks were sectioned into slices of approximately 300 nm thickness using a microtome (PowerTome Ultramicrotome, Boeckeler Instruments). The sections were transferred to coverslips, and the embedding medium was removed with methyl ethyl ketone.

Image analysis

Optical microscopic images were captured in transmission mode using a digital microscope (KH-8700, Hirox, Oradell, NJ, USA). Scanning electron microscopy (SEM) images were obtained using a Phenom G6 ProX SEM (Nanoscience Instruments, Phoenix, AZ, USA) operated at an accelerating voltage of 10 kV. Prior to SEM imaging, samples were sputter-coated with a 5 nm-thick gold layer using a LUXOR sputter coater (Aptco Technologies, Nazareth, Belgium).

XRD experiment

XRD patterns were collected at room temperature using an Empyrean diffractometer (Malvern Panalytical, Malvern, England) with Cu K α radiation (average wavelength = 1.5418 Å) and a nickel filter to reduce

Cu K β contributions. The instrument was operated with a spinning sample holder over a 2θ range of $5\text{--}80^\circ$ with a step size of 0.039° , employing radial divergence slits during measurement. The divergence slits were variable.

The Segal CrI (%) was calculated by the following equation (Segal et al. 1959):

$$\text{CrI} = \frac{I_t - I_a}{I_t} \times 100 \quad (1)$$

where I_t is the intensity of the 200 peak at 22.7° 2θ and, I_a represents the amorphous scattering, determined from the intensity at its lowest value near 18.6° 2θ .

Software

The open-source MAUD software (Materials Analysis Using Diffraction, version 2.9999) was obtained from its official GitHub repository and installed on an HP desktop computer running Microsoft Windows 11. MAUD is a Java-based program originally developed for macOS and is also compatible with other operating systems, including Windows and UNIX/Linux. Several example datasets and tutorials, including those distributed with the MAUD installation and available online, demonstrate Rietveld refinement procedure. However, they provide limited discussion of the practical issues encountered when analyzing weak or poorly resolved cellulose diffraction patterns and in quantifying its crystallinity. For this tutorial, no separate calibration of the instrument was performed. Instrumental calibration can be achieved by measuring a standard reference material, such as NIST LaB $_6$ or silicon, under the same XRD conditions as the samples. The Caglioti and other related parameters are then saved in an instrument file that is used for the samples instead of the default.par file”.

Diffraction patterns of completely periodic crystals of different sizes (as defined by the input FWHM) were calculated using Mercury (version 2025.1.1, Community License) (Macrae et al. 2008). A CuK α weighted average wavelength of $1.5418 \text{ \AA}$ was used, with K β radiation filtered out using a Ni filter. This wavelength was specified directly within Mercury, which retains the setting across sessions. To correct for preferred orientation, the March-Dollase function

was applied for the {001} plane family (Dollase 1986).

Results and discussion

Main inputs to MAUD

Load data file

The XRD pattern was collected from cotton fibers ground into powder using a Wiley mill. Segal et al. (1959) and Nelson and Conrad (1948) reported minimal decrystallization resulting from milling with this type of mill. Optical microscopy shows fiber fragment lengths ranging from approximately $100 \mu\text{m}$ to 1 mm with random orientation (Fig. 2A). The corresponding SEM image reveals the characteristic ribbon-like morphology with convolution twists typical of mature cotton fibers (Fig. 2B). Cross-sectional SEM images (Figs. 2C and 2D) show the expected kidney bean-like profile with average diameters of $21.8 \pm 2.7 \mu\text{m}$ and $9.2 \pm 3.2 \mu\text{m}$ along the longest and shortest axes, respectively. This micrometer-scale structure is built from a hierarchical assembly of elementary fibrils measuring roughly $3\text{--}5 \text{ nm}$ in diameter (Wakelyn et al. 2006). The crystalline signal detected by XRD originates from regions of cellulose that maintain long-range structural order within these nanoscale fibrils. Due to the small fibril diameter, a relatively large fraction of cellulose chains is located at the fibril surface, where their positional coherence is reduced compared to the fibril core because of increased conformational freedom. Consequently, these surface chains adopt a partially ordered, paracrystalline arrangement rather than a fully crystalline state, contributing to the diffuse scattering (Ioelovich et al. 2010, Martínez-Sanz et al. 2017).

The collected XRD pattern was saved as “cotton.txt”, a plain text file (provided in the Supplementary Information) for import into MAUD as a dataset. MAUD provides tools for loading datasets, computing theoretical diffraction patterns, and performing Rietveld refinement via its user-friendly interface (Fig. S1; all such supplementary figures are located in the RefinementProcessGuide.pdf). Input files should be formatted as two columns: 2θ (angle) and intensity. It is recommended to restrict the 2θ range to $10\text{--}50^\circ$, which provides a practical balance between signal

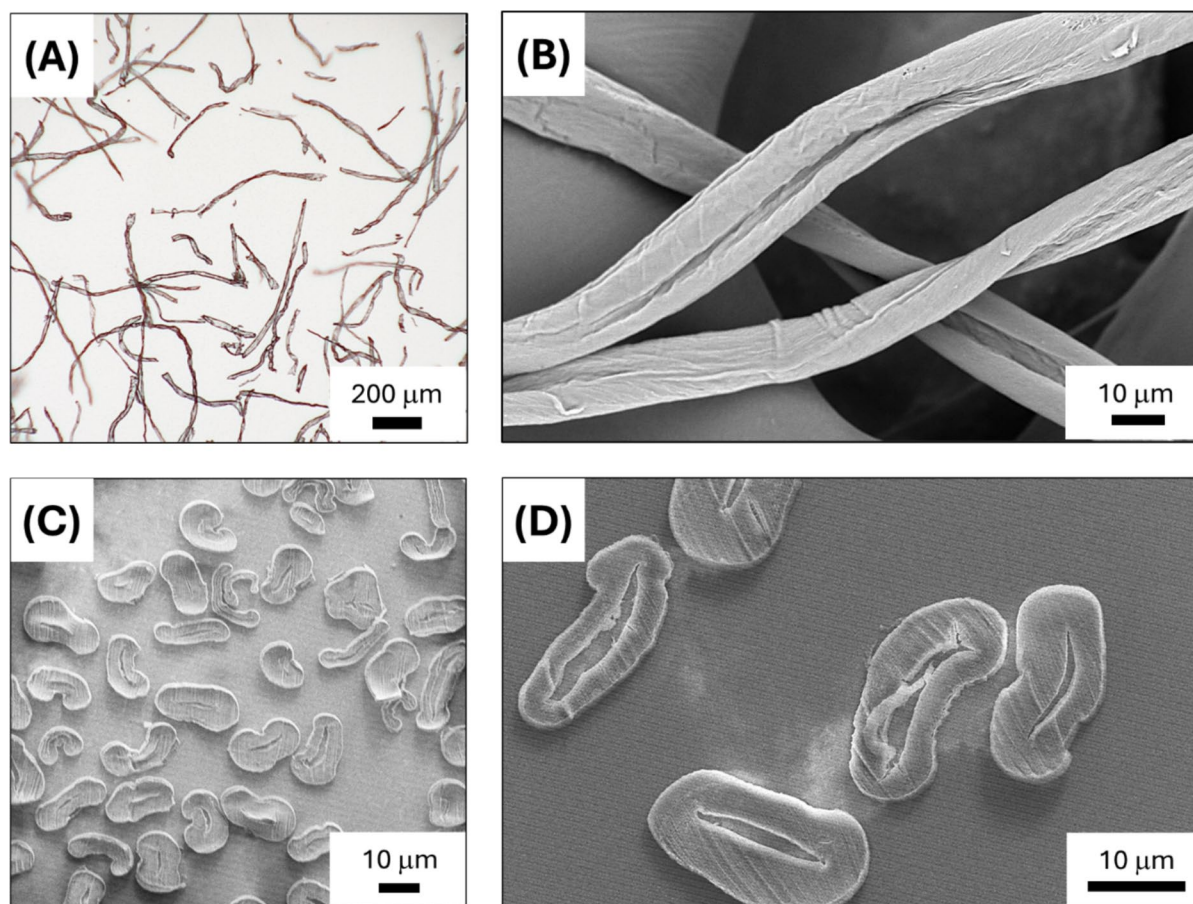


Fig. 2 **A** Optical microscopy image and **B** SEM image of the longitudinal views of ground cotton fibers used for XRD analysis. Corresponding SEM images of fiber cross-sections at **C** low magnification and **D** high magnification

intensity and fitting stability for low-crystallinity cellulose samples. Low-angle regions are often dominated by background, whereas high-angle regions tend to show weaker intensities and higher sensitivity to instrumental and structural model discrepancies (Young 1993; Pecharsky and Zavalij 2003). When raw intensity values exported from the diffractometer are numerically large, the data can be rescaled (e.g., divided by 100) to facilitate data handling within MAUD without altering relative intensities.

Once the dataset is loaded, the diffraction pattern appears in MAUD's main window. Current versions of MAUD also support direct import of Panalytical XRDML files and similar formats from Bruker and Rigaku instruments. To load data, select the "Datasets" tab and choose "Load datafile" from the *File* menu. For visualizing the pattern, users can access

Plot options from the *Graphic* menu and select the "Linear" display mode. By default, MAUD displays diffraction patterns using square root intensity scaling to enhance the visibility of weak peaks. This visualization helps users identify minor reflections that might otherwise be obscured by high-intensity peaks. Selecting "Linear" displays the raw intensities, which is useful for quantitative comparisons or absolute intensity analysis.

Phases

The phase data for cellulose I β is entered by selecting the "Phases" tab in MAUD. Such data are typically provided as .cif files, which are plain text files commonly included in supplementary material of publications or available from online crystallographic

databases. In this study, the crystal structure reported by Nishiyama et al. (2002) was used for cellulose I β . They determined the unit cell dimensions as $a=7.784$ Å, $b=8.201$ Å, $c=10.380$ Å, and $\gamma=96.55^\circ$ using synchrotron X-ray and neutron fiber diffraction data obtained from tunicin cellulose. The corresponding.cif file is provided in the Supporting Information. To import the cellulose I β phase, click the cylinder icon with the left-pointing arrow on the MAUD toolbar. Then the “CelluloseI” will appear at the bottom in the lower region of the MAUD interface (Fig. S2). Additional phases can be added as needed.

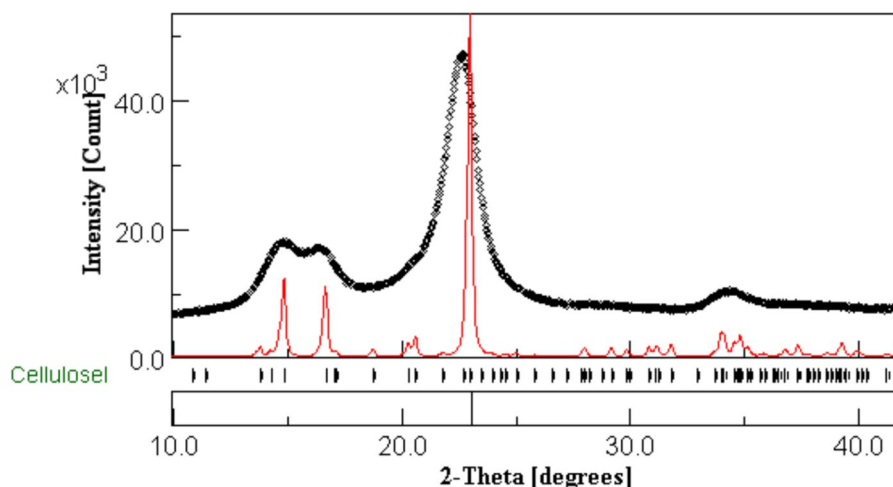
Manual input

Manual input in MAUD enables users to adjust model parameters before running a refinement. To adjust the scale factor, locate the “DataFileSet_x” in the lower region of the MAUD interface. Expanding this node reveals sections including “Diffraction Instrument.” Within this section, locate the parameter “pd_proc_intensity_incident”, which sets the initial scaling of the calculated diffraction pattern relative to the measured data. Adjusting this value (to 2000 for the example file cotton.txt with MAUD 2.9999 or 1000 with MAUD 2.99993) brings the calculated pattern into view for an initial comparison after clicking the calculator button. The cotton.txt file features quite high intensity values, with the consequent lack of visible noise in the data. On a practical note, such high counts are practical for diffractometers with “strip” or “diode array” detectors. Other data will require smaller or even larger values. There is no need for an

exact fit because the next step makes a big change. The optimal value depends on the sample’s diffraction intensity. Once the change is made, click the calculator icon to generate an initial comparison of the observed and calculated diffraction patterns shown in Fig. 3. Alternatively, the pattern can be computed by selecting “Compute spectra” from the *Analysis* menu.

When comparing the initial calculated pattern to the observed pattern, two clear discrepancies are apparent. First, it is notable that the main observed peak from the 200 reflection at 22.7° 2θ is at a lower 2θ value relative to its calculated position. Second, the calculated peaks are considerably narrower, similar to those in the pattern with 0.1° FWHM shown in Fig. 1. These differences suggest that cotton cellulose I β has a larger unit cell a -axis and a smaller crystallite size than the ideal pattern for tunicate, identifying these properties as key candidates for refinement. To explore the effect of modifying these parameters manually, expand the “CelluloseI” and locate the “_cell_length_a” parameter. Adjusting its value from 7.784 to 7.9 Å and clicking the calculator icon increases the a -axis length. Similarly, under the same node, open the “Isotropic” and change “_riet_par_cryst_size” from 1000 to 80 to represent smaller crystallite dimensions. After these manual inputs, the calculated diffraction pattern aligned more closely with the observed data (Fig. S3). However, the overall intensity is, after the decreased crystallite size, lower than that of the observed data, suggesting the need to increase the scale factor to about 10,000 and incorporate an appropriate background function in the model. With the same total number of X-ray photons,

Fig. 3 Observed (black diamonds) and calculated (red line) XRD patterns prior to refinement. The calculated pattern is obtained based on the crystallographic coordinates for tunicin cellulose I β and a crystallite size of 1000 Å



broader reflections (because of smaller peaks) distribute the intensity over a wider range, leading to lower peak heights.

First refinement step: linear background, scale factor, a -axis, and crystallite size

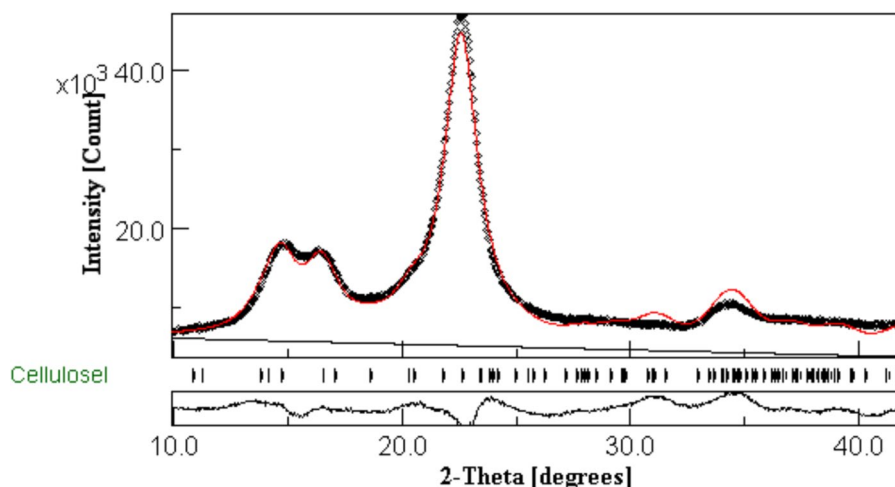
The first refinement step targets five parameters, including linear background, scale factor, a -axis, and crystallite size, to establish a stable starting point for subsequent steps (Fig. S4). The background is the continuous baseline intensity arising from non-sample, environmental sources such as instrument, air, and sample holder scattering. The scale factor adjusts the overall intensity of the calculated diffraction pattern to match the observed pattern. Refinement of the a -axis of the cellulose I β unit cell adjusts peak positions, while the crystallite size determines peak broadening. This sequence stabilizes the refinement and reduces parameter correlations, facilitating reliable adjustments in later steps. To incorporate the linear background function, locate the parameters, “_riet_par_background_pol0,” and “_riet_par_background_pol1” under the “DataFileSet_x,” corresponding to the zero- and first-order polynomial terms, respectively. Change their statuses from “Fixed” to “Refined.” For the scale factor, expand the “Diffraction Instrument” under the “DataFileSet_x” and change the status of “pd_proc_intensity_incident” from “Fixed” to “Refined” with its value set to 1000. To refine the a -axis of cellulose I β , expand the “CelluloseI” and change the status of “_cell_length_a” from “Fixed” to “Refined.” For the crystallite size refinement, expand the “Isotropic”

under the “CelluloseI” and change the status of “_riet_par_cryst_size” from “Fixed” to “Refined” with its value set to 80. After these changes, press the “Launch parameters refinement” button to start the refinement.

To visualize the background, open the “Graphic” menu, select “Plot options”, and check the box labeled “Plot Background”. This displays the background line on the diffraction pattern. Finally, execute the refinement by clicking the “Launch parameters refinement” button (represented by a slot machine or a hammer icon) on the toolbar. It may be necessary to click the “hammer” or “slot machine” button more than once to fully converge the refinement. The process should converge within a few seconds. Figure 4 shows the diffraction pattern after the first refinement step. Overall, the calculated pattern closely matches the observed pattern, indicating the significance of the parameters chosen for refinement in the first refinement step. Some misfit remains, however, near the $1\bar{1}0$, 110 , and 004 reflections at approximately 14.7° , 16.6° , and 34.8° 2θ , respectively.

The MAUD output window displaying the numerical results of the refinement can be expanded by dragging its borders with the system mouse (Fig. S5). The window presents several indicators that assess the agreement between the observed and calculated diffraction patterns. These include “sig”, “Rwp”, and “Rb”, where lower values indicate a better fit. In MAUD, *sig* represents the goodness-of-fit, calculated as the ratio of *Rwp* to *Rexp*, where *Rexp* is the expected R-factor based on counting statistics. *Rwp* (weighted profile R-factor) measures how closely the calculated pattern matches the observed intensities across the profile, while *Rb* (Bragg R-factor) evaluates the agreement between calculated

Fig. 4 Observed (black diamonds) and calculated (red line) XRD patterns after the first refinement step, incorporating five parameters: linear background, scale factor, a -axis, and crystallite size. The fitted linear background line is shown in the pattern



and observed Bragg peak intensities. When these values stabilize across refinement steps, it usually indicates that the refinement has converged and further refinement will bring little improvement. In this first refinement step, the obtained values were $sig=6.71$, $Rwp=6.62$, and $Rb=4.84$.

Second refinement step: quadratic background, b -axis, and γ -angle

In a conventional powder XRD scan, the environmental background typically appears as a smoothly varying curve that is elevated at low angles and gradually decreases with increasing 2θ . The noticeable misfit near the $\bar{1}10$ and 110 reflections after the first refinement step indicates the need to refine the b -axis and γ -angle, which affect their spacing and relative intensities (Fernandes et al. 2011, Altaner et al. 2014). Accordingly, in the second refinement step, the quadratic background, b -axis, γ -angle were added (Fig. S6). The c -axis was left fixed as it represents the chain repeat direction and is less influenced by microstructural changes. To refine the quadratic background, locate the parameter, “_riet_par_background_pol2” under the “DataFileSet_x,” which represents the second-order polynomial term, and change its status from “Fixed” to “Refined.” To refine the b -axis and γ -angle of cellulose I β , expand the “CelluloseI” and change the statuses of “_cell_length_b” and “_cell_angle_gamma” from “Fixed” to “Refined.” After these changes, press the “Launch parameters refinement” button to start the refinement. After the second refinement step, the statistical indicators improved:

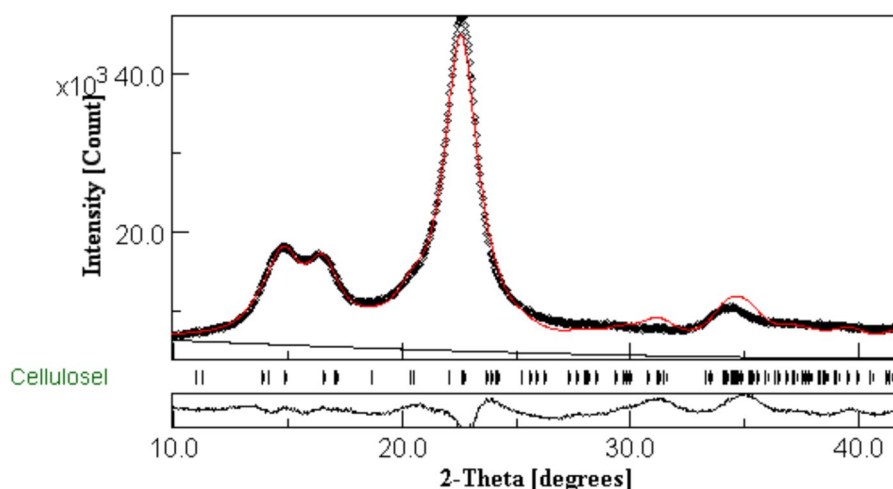
sig decreased from 7.05 to 6.54, Rwp from 6.47 to 5.99, and Rb from 4.94 to 4.32. Figure 5 shows the diffraction pattern after the second refinement step. Consistent with these improvements, the fit near the $\bar{1}10$ and 110 reflections is clearly improved. However, discrepancies persist near the 004 reflection, indicating the need for further refinement.

Third refinement step: preferred orientation

The remaining misfit in the profile of the 004 reflection, which consists of several closely spaced sub-peaks (Fig. 3), is typically influenced by the way fibrous samples settle on the sample holder so that many crystallites become aligned parallel to the sample surface. To account for these preferred orientation effects, MAUD provides (among others) the March-Dollase function (Fig. S7). In MAUD, this is accessed by opening the “Phases” tab in the upper-left window and selecting the “CelluloseI.” Next, click the “eye” icon labeled “Edit the special object of the visible list” located below the “Special” menu at the top. In the new window, go to the “Advanced models” tab, locate the “Texture:” dropdown menu, and select “March Dollase.” Then click “Options,” followed by “Add term” (plus sign) to enable the function. Confirm the changes by selecting OK in both dialog boxes. After this step, the lower portion of the MAUD interface will display “March-Dollase” under “CelluloseI.”

For typical cellulose samples in a diffractometer with Bragg–Brentano geometry, particles with high axial ratios such as fibers prefer to align with the plane

Fig. 5 Observed (black diamonds) and calculated (red line) XRD patterns after the second refinement step, incorporating three additional parameters: quadratic background, b -axis, and γ -angle. The fitted quadratic background line is shown in the pattern



of the sample holder. This preferred orientation results in a plurality of the (001) planes being perpendicular to sample holder and beam, which keeps those fibers from achieving the Bragg condition for diffraction. This diminishes their intensity in the observed diffraction pattern. To model this effect in MAUD (Fig. S8), expand “unknown” under “March-Dollase” and change the values of “_riet_hkl_orient_plane_h” and “_riet_hkl_orient_plane_k” from 1 to 0, while keeping the value of “_riet_hkl_orient_plane_l” at 1. Next, change the status of “_riet_par_pref_or_coef” from “Fixed” to “Refined”. After applying these changes, initiate the refinement by clicking the “Launch parameters refinement” button. This third refinement step reduced *sig* to 4.1, *Rwp* to 3.7, and *Rb* to 3.0. As shown in Fig. 6, introducing the preferred orientation parameter improved the fit of the 004 reflection profile. It is also noticeable that the high 2 θ intensities and the background contribution was diminished.

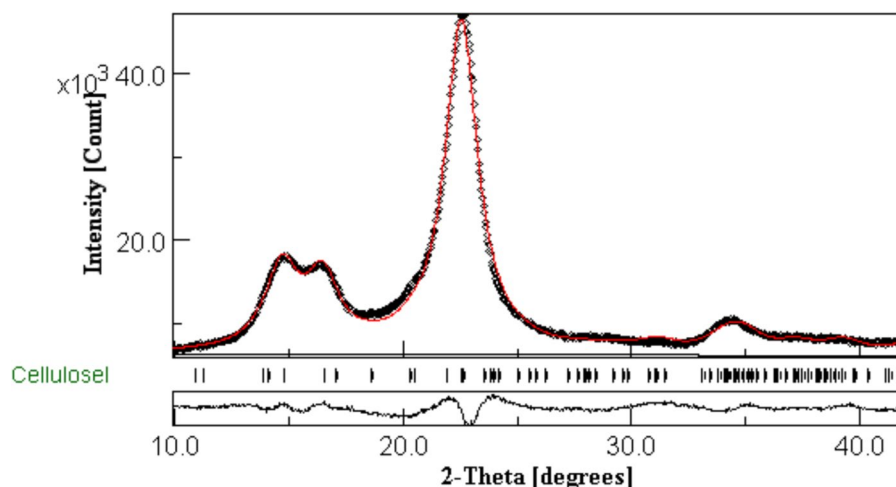
Fourth refinement step: phase composition

Based on the semi-crystalline description of cellulose, the amorphous volume fraction was incorporated in the fourth refinement step. The amorphous contribution was modeled using cellulose amorphized by vibratory ball milling. Under the established milling conditions, the process effectively removes crystalline structure and produces a reproducible amorphous scattering profile (Ling et al. 2019). The ball-milled cellulose provides an experimental approximation of the disordered

signal from amorphous cellulose regions and therefore serves as a practical amorphous reference for quantifying the crystalline contribution. The SEM image of the ball-milled material shows loss of the native fibrous morphology and the formation of aggregated particles (Fig. 7A). The corresponding XRD pattern exhibits a single broad halo centered around 21° 2 θ , characteristic of amorphous cellulose (Fig. 7B). This distinctive scattering profile allows the amorphous contribution to be separated from the slowly varying background. To simulate this amorphous cellulose, the cellulose II crystal structure reported by Langan et al. (2001), with unit cell parameters, $a=8.10$ Å, $b=9.03$ Å, $c=10.31$ Å, and $\gamma=117.10^\circ$ determined from synchrotron XRD of mercerized ramie cellulose II (Its.cif file is provided in the Supporting Information). These values are consistent with those reported for cellulose II prepared by deacetylation of cellulose acetate ($a=8.076$ Å, $b=9.144$ Å, $c=10.386$ Å, and $\gamma=117.100^\circ$). When the crystallite size of cellulose II was set to 12 Å, the calculated diffraction pattern closely matched the experimental diffraction of ball-milled cotton (Nam et al. 2016). Morales et al. (2026) similarly used a small isotropic cellulose I as a model of amorphous. They mentioned that their amorphous model was used by Lutterotti (2012), aligned with that of Le Bail (1995). Our preference is to use the cellulose II model because it better matches the experimental amorphous material (Fig. 7B).

To incorporate phase composition in MAUD, select the “Phases” tab and click the cylinder icon with the left-pointing arrow (Fig. S9). Then, expand

Fig. 6 Observed (black diamonds) and calculated (red line) XRD patterns after the third refinement step, incorporating a preferred orientation parameter modeled with the March-Dollase function



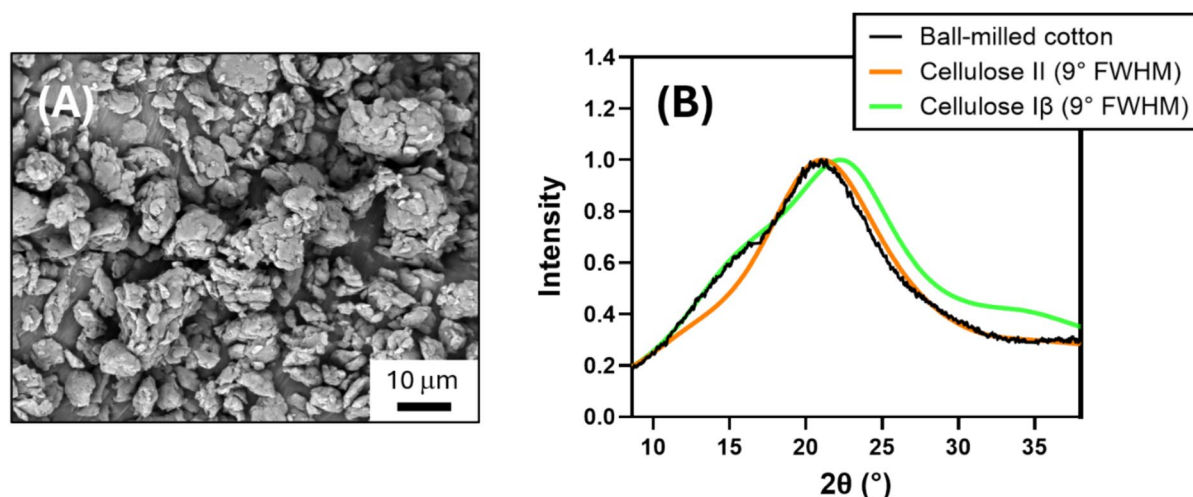


Fig. 7 **A** SEM image of the ball-milled cotton sample. **B** Observed XRD pattern of ball-milled cotton and calculated diffraction patterns of cellulose II and cellulose I β generated from their CIF files using 9° FWHM in Mercury

the “layer 1” under the “default par” and change the statuses of both “Volume fraction of CelluloseI” and “Volume fraction of CelluloseII” from “Fixed” to “Refined” (Fig. S10). Next, under the “CelluloseII,” expand “Isotropic,” set the value of “_riet_par_cryst_size” to 12, and keep its status as “Fixed” (Fig. S10). After applying these settings, click the “Launch parameters refinement” button to start the refinement. This fourth refinement step further reduced *sig* to 3.0, *Rwp* to 2.8, and *Rb* to 2.2. The drop of *Rwp* from 3.7% to 2.8% is substantial, given how close the fit was already. Figure 8 shows the calculated diffraction pattern, demonstrating improved fitting at the point of the indicator of amorphous content in the work of

Segal et al. (1959). Looking back at Fig. 6, one could conclude that some amorphous contribution was needed to “push” the calculated intensity closer to the experimental line.

Rietveld refinement results

Across four refinement steps, a total of eleven parameters were optimized. Table 1 summarizes the refined parameter values, their estimated standard deviations based on the least-square covariance matrix, and figures of merit after the fourth refinement step. The refined *a*-axis value was 7.906 Å, which was larger than the value (7.784 Å) for tunicin cellulose, a highly

Fig. 8 Observed (black diamonds) and calculated (red line) XRD patterns after the fourth refinement step incorporating the volume fractions of crystalline and amorphous celluloses

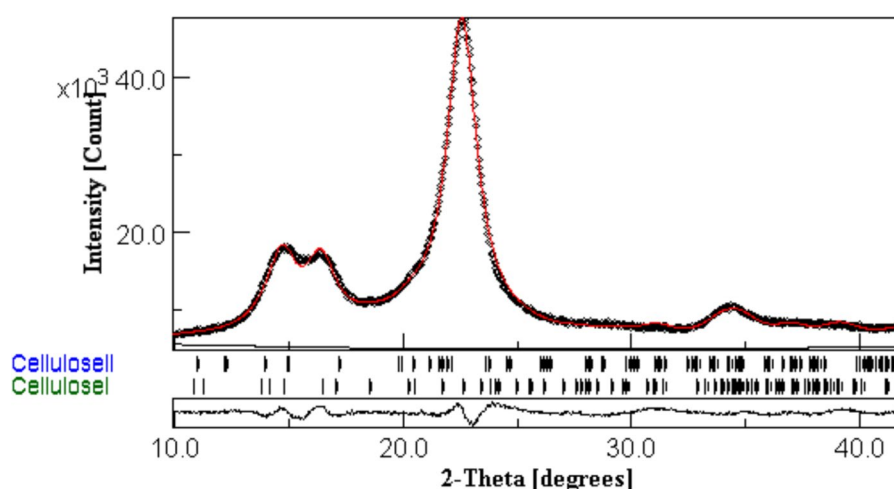
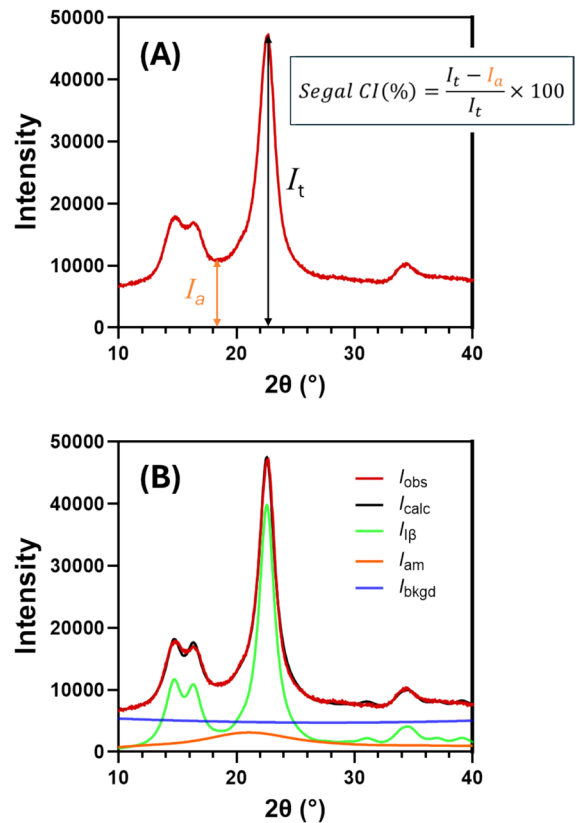


Table 1 Refined parameters, estimated standard deviations, and figures of merit from four Rietveld refinement steps of the experimental XRD pattern of cotton fibers

Parameters and indicators		Refined values	Estimated standard deviation
Scale factor	Intensity	10,100	200
Background	A	6317	85
	B	− 114	9
	C	2.1	0.2
Cellulose I β	a-axis	7.906 Å	0.001
	b-axis	8.220 Å	0.005
	γ	96.16°	0.04
	Crystallite size	66.4 Å	0.4
March-Dollase	Coefficient	1.93	0.03
Volume fraction	Crystalline phase	0.65	6.7
	Amorphous phase	0.35	3.6
Figures of merit	Sig	3.03%	—
	Rwp	2.77%	—
	Rb	2.18%	—

crystalline form derived from sea tunicates (*Halocynthia roretzi*). In contrast, the refined *b*-axis (8.220 Å) and γ -angle (96.157°) were close to those of tunicin cellulose (8.201 Å and 96.55°, respectively). The crystallite size of cotton fibers was determined to be 66.35 Å. The March-Dollase coefficient refined to 1.925, indicating a substantial deviation from the value of 1 expected for a randomly orientated specimen. A coefficient greater than 1.0 implies a decrease in the fraction of *hkl* planes that can meet the Bragg condition for reflection. This value represents an apparent correction derived from a single diffraction pattern rather than determination of the true orientation distribution. Obtaining the actual orientation distribution would require diffraction data collected over multiple sample tilts or geometries, which enable reconstruction of a full orientation profile (Morales et al. 2026). The volume fractions of crystalline and amorphous phases refined to 0.65 and 0.35, respectively, yielding a crystallinity of 65%. Their estimated standard deviation values, which were derived from the covariance matrix of a single refinement, are large, reflecting high parameter correlations due to overlapping diffraction features in cellulose. However, repeated refinements under different initial conditions

**Fig. 9** **A** Measurement of Segal crystallinity index (CrI) of cotton fibers from two peak intensities at 22.7° 2 θ and 18° 2 θ in the observed diffraction pattern (Eq. 1). **B** Diffraction intensity profiles as a function of 2 θ for the observed pattern (I_{obs}), calculated pattern (I_{calc}), and refined phase contributions including cellulose I β (I_{β}), amorphous cellulose (I_{am}), and background (I_{bkgd})

yielded consistent phase fractions (0.650 ± 0.001 for crystalline cellulose), indicating the stability of the solution within the chosen model. The determined crystallinity was much lower than the Segal CrI of 77% calculated from the height of the 200 peak and the intensity minimum at about 18.6° (Fig. 9A), supporting the recognized tendency of Segal's method to overestimate CrI (Nam et al. 2024a, b). The overestimation is attributed to its simplified, peak-height-based approach, which does not account for the full diffraction profile or overlapping peaks (French and Santiago Cintrón 2013; French 2022). Moreover, the Segal method fails to capture the maximum scattering of the amorphous cellulose, which occurs at around 21° 2 θ (Fig. 7B). Instead, it measures intensity at 18° 2 θ (Eq. 1).

The intensities as a function of 2θ for all refined pattern contributions can be exported to a spreadsheet. To do this, open the *Graphic* menu on the toolbar and select “Plot selected dataset” to launch the auxiliary MAUD window. Within this window, go to the *Tools* menu and choose “Export experimental/computed data” to save the data file (Fig. S11). Figure 9B presents the intensity profiles as a function of 2θ for observed pattern, calculated pattern, and refined phase contributions, including cellulose I β , amorphous cellulose, and background.

Figure 10 shows the variation in refinement quality across successive refinement steps. As additional parameters were introduced, the goodness-of-fit indicator (*sig*), and residual errors (*Rwp* and *Rb*) progressively decreased, reflecting improved agreement between the observed and calculated diffraction patterns and successful model convergence. The consistent downward trend supports the inclusion of the parameters across the four refinement steps to achieve a reliable fit.

Limitations and scope

Although this workflow provides a structured approach for quantifying cellulose crystallinity using Rietveld refinement, several limitations should be considered. (a) Two-phase representation. The method models semi-crystalline cellulose as a

two-component system consisting of crystalline and amorphous cellulose phases. This simplified representation does not explicitly include paracrystalline domains. b) Choice of amorphous reference. The workflow assumes that the scattering profile of ball-milled cellulose adequately represents the amorphous cellulose and does not include non-cellulosic amorphous components. (c) Background-amorphous separation. The background signal is treated as a smooth function, whereas amorphous cellulose produces a broad, non-periodic halo. The separation of these distinct intensity profiles is not unique and depends on the characteristics of the background. (d) Material-specific considerations. The protocol is optimized for typical cellulose-rich materials. Materials with mixed cellulose polymorphs, pronounced preferred orientation, significant non-cellulosic phases may require supplemental modeling. Further limitations are the lack of tailoring of the MAUD software for the explicit diffractometer. The crystal information file resulted in an *R_w* based on *F*² of 42% so there must be some error in the atomic positions in the model structure. Some indicators of the quality of the study have not been discussed, needed especially when more advanced variables are considered. The order of adding parameters has not been discussed beyond the selection of the first five. Perhaps the amorphous could be introduced third, instead of March-Dollase function. If the crystalline material has preferred 001 orientation, then perhaps the amorphous material does also.

Conclusions

To overcome limitations in cellulose crystallinity measurement using traditional XRD methods and to encourage more comprehensive analysis of powder diffraction data, we present a cellulose-specific workflow for quantifying true crystallinity of cellulose using Rietveld refinement with the open-source MAUD software. The refinement process was structured into four sequential steps, each designed to account for cellulose-specific structural characteristics. Ten key parameters—including scale factor, background function, unit cell geometry, crystallite size, preferred orientation, and phase composition—were systematically refined to represent the two-phase nature of cellulose fibers. The strong agreement

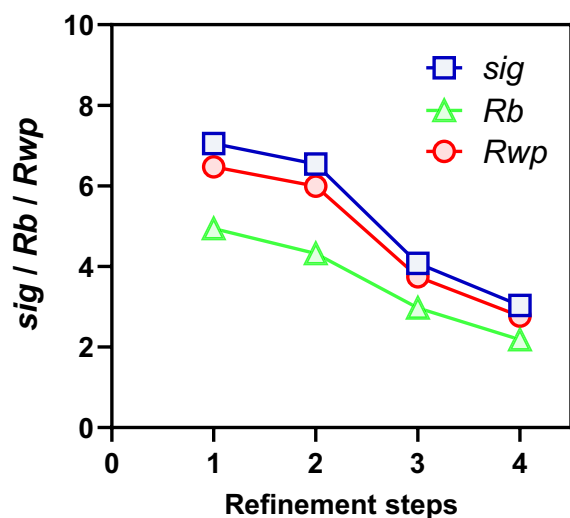


Fig. 10 Change in goodness-of-fit (*sig*), and residual errors (*Rb* and *Rwp*) over successive refinement steps

between observed and calculated diffraction patterns reflected in the refinement-specific figures of merit, supports the reliability and robustness of this approach.

Designed for accessibility, this protocol lowers the barrier for researchers without prior experience in crystallographic refinement to perform crystallinity measurements using the entire one-dimensional powder diffraction profile analysis. Future work could explore additional refinement procedures, such as introducing an additional March-Dollase term to account for more complex orientation distributions, refining anisotropic crystallite sizes to capture directional size effects, or expanding the amorphous model to include possible alignment contributions, and examining how these refinements influence crystallinity measurements. Another potential direction is the application of this workflow to cellulose reference materials across different diffractometers and user groups to help establish more robust reference crystallinity values. A key outstanding question is whether the number of unique diffraction data points is sufficient to reliably support the number of refined variables. This challenge is particularly relevant for cellulose samples, where broad peaks, overlapping reflections, and low crystallinity often limit the amount of independent structural information available. Indicators such as nonphysical parameter values or unit cell dimensions outside reasonable limits can provide a practical check on refinement reliability.

Supplementary information

Six individual files are provided to assist users with the cellulose-specific Rietveld refinement method:

- (1) One .txt file: Supplementary Information file 1.txt—XRD data of cotton fiber formatted for import into MAUD as a dataset.
- (2) Two .cif files: Supplementary Information file 2.cif and Supplementary Information file 3.cif—crystal information files representing the cellulose I β and cellulose II phases, respectively, for use in MAUD. The Supplementary Information file 2.cif file is based on the neutron diffraction study of Nishiyama et al. (2002) that includes the positions of the hydroxy hydrogen (deuterium) atoms, including the observed fractional occupancies. Those atoms are not available from the

synchrotron X-ray results. The Supplementary Information file 3.cif file was forwarded by Paul Langan to ADF. Although based on Langan et al. (2001), it also includes hydrogen atom positions that are not available in that work. Both files were part of the Supplementary Information in French (2014) in a combined file that is not directly useful in MAUD.

- (3) One .par file: Supplementary Information file 4.par—an analysis file containing the complete analysis used in this study. It can be opened in the MAUD software by selecting “Open analysis” from the File menu.
- (4) One .pdf file: Supplementary Information file 5.pdf—a document containing screenshots of the MAUD interface, illustrating the sequential steps of the refinement procedure.
- (5) One .xlsx file: Supplementary Information file 6.xlsx—spreadsheet containing refinement results exported from MAUD (“Export experimental/calculated data”).

The .txt, .par, and .cif files can be viewed with simple text-editing software such as Windows Notepad or Mac TextEdit.

Acknowledgement Mention of trade names or commercial products is solely for the purpose of providing specific information and does not imply recommendation or endorsement by USDA. USDA is an equal opportunity provider and employer.

Author contributions Sunghyun Nam: Data curation, Analysis, Investigation, Writing-Original draft, Visualization; Alfred D. French: Conceptualization, Methodology, Writing-Review & Editing; Md Nayeem Hasan Kashem: Analysis, Writing-Review & Editing.

Funding This research was supported by the U.S. Department of Agriculture, Agricultural Research Service.

Data availability Data is provided within the manuscript and supplementary information files.

Declarations

Conflict of interest The authors declare no conflict of interests.

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